

Project Title

Building a Chatbot using LangChain for Querying Information in Vietnamese Legal Documents

1. Project Overview

This project builds a chatbot that can answer questions based on Vietnamese legal documents. The system uses LangChain to connect the language model, embeddings, and vector database. Users can type a legal question, and the chatbot will search for relevant information from the documents and generate an answer.

2. Tools and Libraries

The main tools used in this project are:

- Python
- Google Colab
- LangChain
- Gemini API
- ChromaDB
- Gradio
- Pandas
- BERTScore

3. Environment Setup

First, open the notebook in Google Colab.

Then, install the required libraries:

```
!pip install langchain langchain-community langchain-google-genai  
langchain-chroma chromadb gradio bert-score openpyxl python-dotenv
```

After installing the libraries, restart the runtime if needed.

4. API Key Setup

This project uses Gemini API. Therefore, you need to prepare a Gemini API key.

When the notebook asks for the API key, enter your key:

```
import os
```

```
os.environ["GEMINI_API_KEY"] = input("Enter Gemini API key: ")
```

5. Upload Data Files

The system needs two text files:

1. A legal document file
2. A question-answer file

Upload the legal document file first. This file contains the content of Vietnamese legal documents.

Then, upload the QA file. This file contains sample questions and answers.

6. Data Processing

After uploading the files, the notebook will clean and process the text data. The legal documents and QA data are converted into document objects so that they can be used by LangChain.

The system will also split and organize the documents before storing them in the vector database.

7. Create Embeddings

The project uses Gemini Embedding to convert text into vectors.

```
from langchain_google_genai import GoogleGenerativeAIEmbeddings
```

```
embedding_model = GoogleGenerativeAIEmbeddings(  
    model="models/gemini-embedding-001"  
)
```

These vectors help the system find relevant information based on the user's question.

8. Create Vector Database

The project uses ChromaDB as the vector database.

```
from langchain_chroma import Chroma
```

The processed documents are stored in ChromaDB. When a user asks a question, the system searches this database to find the most relevant documents.

9. Create the Language Model

The chatbot uses Gemini as the language model.

```
from langchain_google_genai import ChatGoogleGenerativeAI
```

```
llm = ChatGoogleGenerativeAI(  
    model="gemini-2.5-flash",  
    temperature=0  
)
```

The model generates the final answer based on the retrieved legal information.

10. Run the Chatbot

The chatbot interface is built with Gradio.

```
demo.launch(share=True)
```

After running this cell, Gradio will create a link. Open the link to test the chatbot.

11. How the System Works

The system works in the following steps:

1. The user enters a legal question.
2. The question is converted into an embedding.
3. The system searches for relevant legal documents in ChromaDB.
4. The retrieved documents are sent to the language model.
5. The chatbot generates an answer based on the retrieved information.
6. The answer is shown to the user through the Gradio interface.

12. Expected Output

The output of the system includes:

- The user's question
- The chatbot's answer
- The retrieved legal context

- The source information
- The response time

13. Notes

Before running the project, make sure that:

- All required libraries are installed.
- The Gemini API key is valid.
- The legal document file is uploaded correctly.
- The QA file is uploaded correctly.
- The data format is correct.
- The Gradio link is working.

14. Common Errors

If the system cannot find any documents, check the format of the input files.

If there is an API error, check the Gemini API key or wait and run the cell again.

If Gradio does not create a link, restart the runtime and run the notebook again.

15. Summary

In summary, this project uses LangChain, Gemini, ChromaDB, and Gradio to build a chatbot for Vietnamese legal documents. The chatbot can retrieve relevant legal information and generate simple answers for users.